

Vegetable intake is associated with reduced breast cancer recurrence in tamoxifen users: a secondary analysis from the Women's Healthy Eating and L...



The protective effect of vegetables on the risk of breast cancer recurrence is uncertain. We sought to evaluate the association between breast cancer recurrence and vegetable intake including analyses stratified on tamoxifen use. Experimental evidence of anti-carcinogenic activity of phytochemicals in cruciferous vegetables in combination with tamoxifen led to specific evaluation of this class of vegetables as well. To assess the association between vegetable intake and breast cancer recurrence, vegetable intake from repeat 24-h dietary recalls were examined as a secondary analysis of 3,080 breast cancer survivors enrolled in the Women's Healthy Eating and Living (WHEL) Study. At the time of enrollment women were, on average, 23.5 months post-diagnosis. The hazard of recurrence, controlling for relevant and significant clinical and demographic variables, with vegetable intake was assessed overall and separately for women taking tamoxifen. WHEL participants reported mean baseline intakes (mean, SE) of 3.1 ± 0.05 and 0.5 ± 0.02 servings/day of total and cruciferous vegetables, respectively. Baseline vegetable intake in the highest as compared to lowest tertiles was associated with an overall lower adjusted hazard ratios (HR) for recurrence of 0.69, 95% CI 0.55-0.87. Among women taking tamoxifen, the HRs were 0.56, 95% CI 0.41-0.77 for total vegetables and 0.65, 95% CI 0.47-0.89 for cruciferous vegetable intake. The hazard in women using tamoxifen who reported cruciferous vegetable intake above the median and who were within the highest tertile of total vegetable intake was HR 0.48; 95% CI 0.32-0.70. This secondary analysis in over 3,000 breast cancer survivors suggests that baseline vegetable intake may be associated with a reduction in the risk of breast cancer recurrent or new events particularly for those using tamoxifen. Such associations should be explored further as the possibility that vegetable intake is simply a surrogate for other health-promoting behaviors cannot be ruled out.